Sales Data Analysis with SQL (SQL Server)

A set of T-SQL queries that analyze sales performance over time, product performance, category contribution, and customer behavior using window functions, CTEs, and conditional segmentation.

Business Questions

- How do sales, customers, and quantity change month over month and year over year?
- What is the cumulative sales trend?
- Which products are growing or declining compared with the previous year and with their own average?
- Which categories contribute the most to total sales?
- How are products distributed across cost ranges?
- How many customers are VIP, Regular, or New?

Data Model

A star schema with three tables:

Table	Type	Key columns
`fact_sales`	Fact	order_number, order_date, product_key, customer_key, sales_amount, quantity, price
`dim_customers`	Dimension	customer_key, first_name, last_name, country, birthdate
`dim_products`	Dimension	product_key, product_id, product_name, category, cost

Tools

Microsoft SQL Server, T-SQL, SSMS

Analyses and SQL Techniques

#	Analysis	Techniques
1	Change over time (monthly sales, customers, quantity)	`YEAR()`, `MONTH()`, `DATETRUNC()`, `FORMAT()`, `GROUP BY`
2	Cumulative analysis (running total and running average)	Window functions: `SUM() OVER`, `AVG() OVER`, subquery
3	Performance analysis (year over year and vs average per product)	CTE, `LAG()`, `PARTITION BY`, `CASE WHEN`
4	Part-to-whole (category share of total sales)	CTE, `SUM() OVER ()`, percentage calculation
5	Product segmentation by cost range	`CASE WHEN`, aggregation
6	Customer segmentation (VIP / Regular / New)	CTE, `DATEDIFF()`, `CASE WHEN`
7	Customer base query (core report columns)	`JOIN`, `CONCAT()`, age calculation

Customer Segmentation Rules

- **VIP:** at least 12 months of history and total spending above 5,000
- **Regular:** at least 12 months of history and total spending of 5,000 or less
- **New:** less than 12 months of history

Sample Query

```sql
```

```sql
-- Yearly sales per product vs previous year
WITH yearly_product_sales AS (
    SELECT
        YEAR(s.order_date)  AS order_year,
        p.product_name,
        SUM(s.sales_amount) AS current_sales
    FROM fact_sales s
    INNER JOIN dim_products p ON p.product_key = s.product_key
    WHERE s.order_date IS NOT NULL
    GROUP BY YEAR(s.order_date), p.product_name
)
SELECT
    order_year,
    product_name,
    current_sales,
    LAG(current_sales) OVER (PARTITION BY product_name ORDER BY order_year) AS pre
FROM yearly_product_sales;
```

Key Findings

> Fill these in from your query results, with real numbers. Examples of the format:

- Sales peaked in **[month/year]** with **[amount]**, and grew **[x%]** year over year.
- **[Category]** contributes **[x%]** of total sales.
- **[x%]** of customers are VIP, and they generate **[x%]** of revenue.
- **[n]** products sold below their own average in the latest year.

Files

- `sales_analysis_queries.sql` : all queries, commented and organized by topic